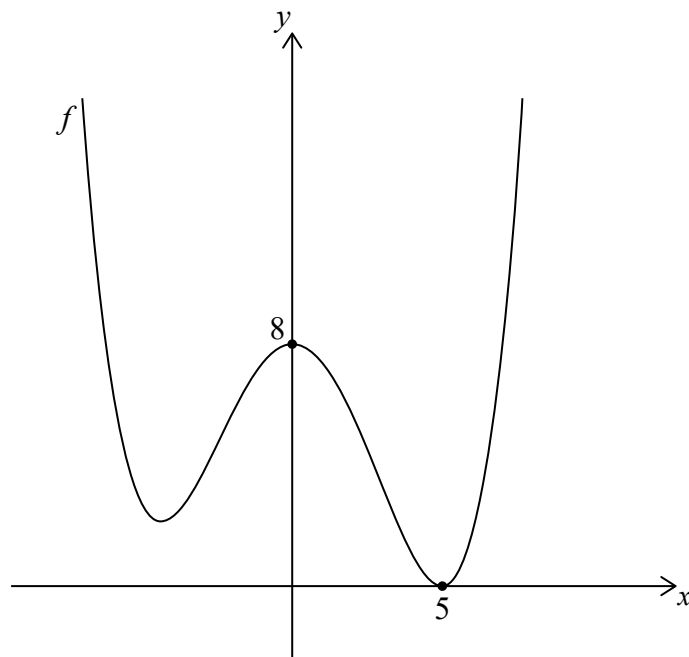


11.3 Classwork: Motion

4. [Maximum mark: 7]

The following diagram shows part of the graph of f with x -intercept $(5, 0)$ and y -intercept $(0, 8)$.



(a) Find the y -intercept of the graph of

(i) $f(x) + 3$;

(ii) $f(4x)$.

[3]

(b) Find the x -intercept of the graph of $f(2x)$.

[2]

(c) Describe the transformation $f(x + 1)$.

[2]

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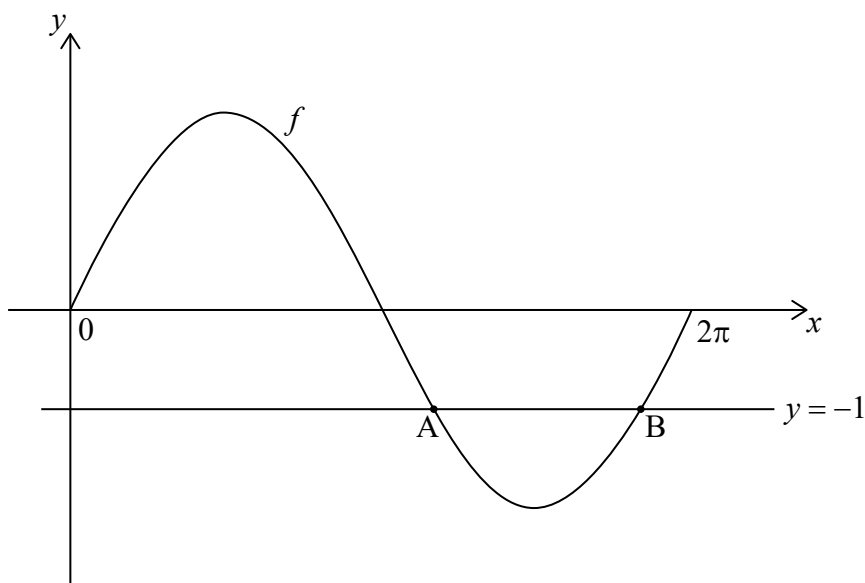
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7. [Maximum mark: 7]

Consider the graph of the function $f(x) = 2 \sin x$, $0 \leq x < 2\pi$. The graph of f intersects the line $y = -1$ exactly twice, at point A and point B. This is shown in the following diagram.



- (a) Find the x -coordinate of A and of B. [4]

Consider the graph of $g(x) = 2 \sin px$, $0 \leq x < 2\pi$, where $p > 0$.

- (b) Find the greatest value of p such that the graph of g does not intersect the line $y = -1$. [3]

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Do **not** write solutions on this page.

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

8. [Maximum mark: 13]

Let $f(x) = 2 \sin(3x) + 4$ for $x \in \mathbb{R}$.

(a) The range of f is $k \leq f(x) \leq m$. Find k and m . [3]

Let $g(x) = 5f(2x)$.

(b) Find the range of g . [2]

The function g can be written in the form $g(x) = 10 \sin(bx) + c$.

(c) (i) Find the value of b and of c .

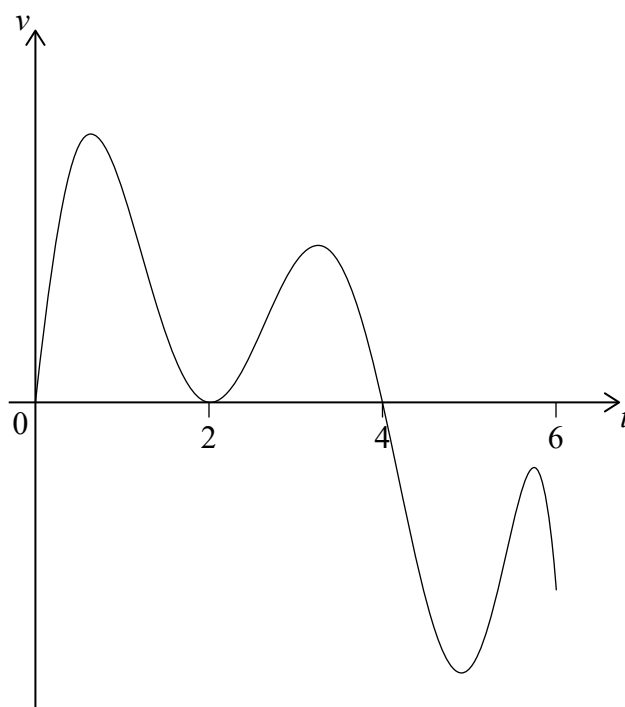
(ii) Find the period of g . [5]

(d) The equation $g(x) = 12$ has two solutions where $\pi \leq x \leq \frac{4\pi}{3}$. Find both solutions. [3]



7. [Maximum mark: 7]

A particle P starts from point O and moves along a straight line. The graph of its velocity, $v \text{ ms}^{-1}$ after t seconds, for $0 \leq t \leq 6$, is shown in the following diagram.



The graph of v has t -intercepts when $t = 0, 2$ and 4 .

The function $s(t)$ represents the displacement of P from O after t seconds.

It is known that P travels a distance of 15 metres in the first 2 seconds. It is also known that $s(2) = s(5)$ and $\int_2^4 v \, dt = 9$.

- (a) Find the value of $s(4) - s(2)$. [2]
- (b) Find the total distance travelled in the first 5 seconds. [5]

(This question continues on the following page)



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Paper 2: Use a calculator

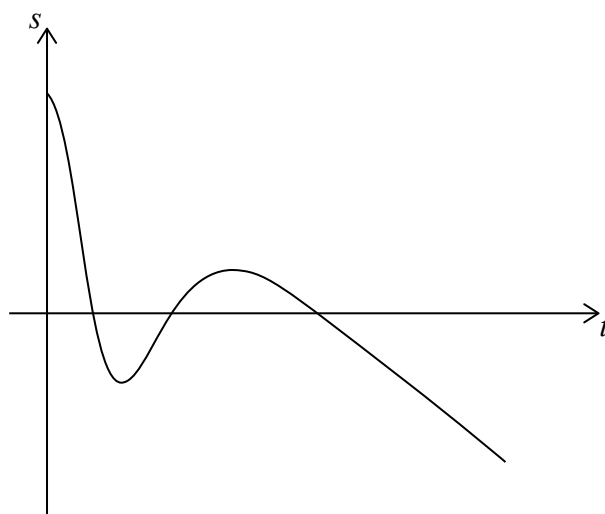
Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

8. [Maximum mark: 16]

In this question distance is in centimetres and time is in seconds.

Particle A is moving along a straight line such that its displacement from a point P, after t seconds, is given by $s_A = 15 - t - 6t^3e^{-0.8t}$, $0 \leq t \leq 25$. This is shown in the following diagram.



- (a) Find the initial displacement of particle A from point P. [2]
- (b) Find the value of t when particle A first reaches point P. [2]
- (c) Find the value of t when particle A first changes direction. [2]
- (d) Find the total distance travelled by particle A in the first 3 seconds. [3]

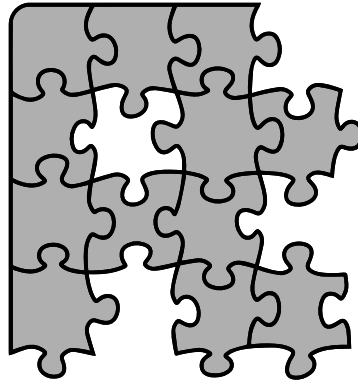
Another particle, B, moves along the same line, starting at the same time as particle A. The velocity of particle B is given by $v_B = 8 - 2t$, $0 \leq t \leq 25$.

- (e) (i) Given that particles A and B start at the same point, find the displacement function s_B for particle B.
- (ii) Find the other value of t when particles A and B meet. [7]



5. [Maximum mark: 6]

A jigsaw puzzle consists of many differently shaped pieces that fit together to form a picture.



Jill is doing a 1000-piece jigsaw puzzle. She started by sorting the edge pieces from the interior pieces. Six times she stopped and counted how many of each type she had found. The following table indicates this information.

Edge pieces (x)	16	31	39	55	84	115
Interior pieces (y)	89	239	297	402	580	802

Jill models the relationship between these variables using the regression equation $y = ax + b$.

- (a) Write down the value of a and of b . [3]
- (b) Use the model to predict how many edge pieces she had found when she had sorted a **total** of 750 pieces. [3]

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